SYEDA HAFSA

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SUMMARY

Computer and Information Systems Engineer and Master of Data Science student at Universiti Kebangsaan Malaysia, turning data and systems into decisions. Built and evaluated machine learning models with repeated cross-validation and statistical tests, analysed data in Python, R and Power BI, and trained a YOLO computer vision model; three internships across data analytics, IT systems and public-sector IT.

EDUCATION

Master of Data Science, Universiti Kebangsaan Malaysia (UKM) *Sep 2026 to present*

Bachelor of Engineering (BE) in Computer and Information Systems Engineering *2021 to 2025*
NED University of Engineering and Technology · CGPA 3.25

HSC (Pre-Engineering), Bahria College · 83.09% (A1) *2019 to 2021*

SSC (Science), St. Joseph's Convent School · 83.76% (A1) *2017 to 2019*

SELECTED PROJECTS

Loan Default Prediction | WEKA 3.8, Python, scikit-learn, imbalanced-learn, Power BI *2026*

- Preprocessed a 255,347-loan dataset in WEKA, Python and Power BI: corrected 17,274 inconsistent records, engineered ratio features, discretised and normalised attributes, and balanced the 11.6% default class with SMOTE.
- Compared Decision Tree, Naïve Bayes, MLP, Random Forest and AdaBoost in a leakage-free cross-validation pipeline with tuning and Friedman/Wilcoxon tests; tuned AdaBoost ranked first (test AUC 0.740, recall 0.659).

Hotel Booking Demand: Statistical Analysis in R | R 4.6.1, tidyverse, ggplot2 *2026*

- Analysed 119,390 hotel bookings in R: uncovered ~129,000 hidden missing values stored as the text "NULL", documented every cleaning decision, and removed 1.7% of rows with justification.
- Applied Welch t-test, ANOVA with Tukey post-hoc, chi-square and multiple regression with effect sizes (d, η^2, Cramér's V, R^2); adding season and a hotel × month interaction lifted explained variance in room rate from 22% to 55%, and wrote a six-point executive memo.

Real-Time Anomaly Detection for CCTV (final year team project) | YOLO11n, OpenCV, Flask, Firebase, CVAT *2025*

- Trained the team's fall and jump detection model (YOLO11n, 7 classes, 15,000+ images, 100 epochs): 0.716 recall and 0.583 mAP50 on validation.
- Annotated video data in CVAT and connected Firebase for live, timestamped event logging. Co-authored a research paper on the system for a NED University journal.

P2P Lending Risk Analysis (Technocolabs internship) | Python, pandas, Power BI

- Explored 113,937 Prosper loans: 17,010 (14.9%) were charged off, defaulted or past due; showed rate, APR and lender yield act as one pricing signal ($r \geq 0.989$); built Power BI dashboards on borrower risk and loan performance.

More projects: Sales Prediction (Random Forest, R^2 0.981) · Car Price Prediction (R^2 0.964) · Multiple Disease Prediction (Streamlit; heart model recall 0.906) · Virtual Mouse (MediaPipe hand tracking, 7 gestures)

EXPERIENCE

Data Analyst Intern, Technocolabs Software *Mar 2026 to Jun 2026*

- Cleaned and preprocessed raw datasets and performed exploratory data analysis to identify trends and data quality issues.
- Built interactive Power BI dashboards for key business metrics; authored performance reports and defined KPIs.

IT Intern, Karachi Development Authority (KDA) *Mar 2024 to May 2024*

- Worked with the authority's existing database in a large public-sector IT department, learning how systems are documented and work is reported upward.
- Prepared and delivered a presentation on working in a government IT environment to a non-technical audience.

IT/IS Intern, Getz Pharma *Aug 2023 to Nov 2023*

- Designed and optimised interactive dashboards to improve data visualisation and reporting efficiency.
- Authored a User Requirement Specification (URS) in collaboration with cross-functional teams.

FEATURED ARTICLES

From Pixels to Patterns: Exploring Connected Regions Using MATLAB, LinkedIn, May 2025

Understanding Morphological Operations in Computer Vision, LinkedIn, May 2025

Can You Teach a Machine to See? Exploring Edge Detection in MATLAB, LinkedIn, May 2025

SKILLS

Data and analytics: Python (pandas, NumPy), R (tidyverse, ggplot2), SQL, Power BI, WEKA 3.8, hypothesis testing (t-test, ANOVA, chi-square, regression), EDA, Matplotlib, Seaborn

Machine learning and vision: scikit-learn, imbalanced-learn (SMOTE), cross-validation, ROC/PR, Friedman and Wilcoxon tests, YOLO (Ultralytics), OpenCV, MediaPipe, CVAT, MATLAB image processing

Software and systems: Streamlit, Flask, Firebase, HTML, CSS, JavaScript, Git and GitHub, Jupyter, Colab, RStudio

ACTIVITIES AND LANGUAGES

Team Leader, Newspaper Editorial Team, NEDUET (2024 to 2025): led a student team creating and publishing content for university events. **Registrations and HR Team Member**, AI Club, NEDUET (2021 to 2023). **Languages:** English (fluent), Urdu (native), Turkish (elementary)